KARNA

AI-Based Environmental Sound Awareness
with Simulated Haptic Alerts

FUNCTIONAL MVP SUBMISSION
Sound-to-alert workflow demonstrated through software and simulation.

Team Details

EVENT	CODEQUEST
THEME	Accessibility + TinyML
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DEMO LINK	Demo Link
GITHUB URL	GitHub Repo

Prototype status

ThecurrentMVP is functional in software and simulation. It accepts audio, runs sound classification, displays the predicted sound and confidence, maps the detected class to a pulse pattern, and demonstrates the pulse count through a Wokwi LED simulation.

Important scope note

TheLED isonlyasimulationaid used to demonstrate the pulse sequence. It is not the intended product. The proposed end product is a wearable haptic wristband; no physical wristband or vibration motor is connected in this MVP.

1. Project Overview

KARNA is an environmental sound awareness system intended to help deaf and hard-of-hearing users become aware of important sounds around them. The prototype converts an audio input into a machine-learning prediction and then translates the detected sound category into a corresponding haptic pulse pattern.

The current MVP demonstrates this complete software workflow. Audio can be supplied either as an uploaded audio file or by recording through the laptop microphone. The predicted sound and confidence are displayed in the Streamlit frontend. The mapped pulse sequence is then demonstrated using a Wokwi simulation in which an LED blinks once for each vibration pulse.

2. Problem and Proposed Approach

People who are deaf or hard of hearing may not notice important environmental or emergency sounds through hearing alone. KARNA explores an alternative alert channel: classify the sound and represent it through a tactile pattern that could be delivered by a wearable wristband.

The prototype follows a sound-to-alert approach:

Audio input	ML inference	Pulse mapping	Wokwi demonstration
Upload a file or record through the laptop microphone	Process the audio and predict a sound class	Translate the predicted class into its configured pulse pattern	Blink an LED once per pulse to represent the haptic output

3. Functional Features Demonstrated

Feature	What the MVP demonstrates
Audio file input	The user can provide a recorded audio file for classification.
Microphone recording	The Streamlit frontend supports recording audio through the laptop microphone.
Machine-learning classification	The audio is processed by the trained KARNA sound-classification model.
Prediction display	The frontend displays the predicted sound category and its confidence.
Haptic pulse mapping	The predicted category is passed through the pulse-mapping logic to select the corresponding pulse pattern.
Wokwi LED simulation	The mapped pulse sequence is demonstrated in Wokwi. The LED blinks as many times as the vibration pattern contains pulses.

4. Technical Implementation

4.1 Audio processing and model

TheKARNA inference pipeline usesaSmallCNN-based sound classifier. The model uses convolutional layers followed by pooling and a final classification layer. The project has mono and stereo inference variants.

The preprocessing pipeline loads WAV audio, handles mono or stereo input, resamples audio to 16 kHz, pads or trims it to 3 seconds, and computes a 64-bin log-Mel representation using an FFT size of 1,024 and hop length of 256. The resulting representation is passed to the trained model for prediction.

In the current MVP, inference runs in software on a laptop. Running the model on the ESP32 for fully on-device operation is planned for the next development stage.

4.2 Pulse mapping and simulation

Afterinference, thepredicted sound class is passed to the pulse-mapping stage. The mapping associates a sound category with a corresponding pulse sequence. The Wokwi simulation represents each pulse with an LED blink, allowing the alert sequence to be observed without a physical vibration motor.
The pulse pattern configured for each sound class is listed below. The Wokwi LED blinks once per pulse for the stated pulse duration, with the stated pause between consecutive pulses (no pause after the last pulse).

Sound class	Pattern ID	Pulses	Pulse duration (ms)	Pause (ms)	Total duration (ms)
ambulance_siren	P01	2	600	200	1400
fire_truck_siren	P02	3	500	200	1900
traffic_noise	P03	1	150	0	150
short_horn_ahead	P04	1	200	0	200
short_horn_behind	P05	2	200	150	550
short_horn_alongside	P06	3	150	100	650
long_horn_ahead	P07	1	1000	0	1000
long_horn_behind	P08	2	800	250	1850
long_horn_alongside	P09	3	700	200	2500
varied_horn_ahead	P10	3	300	150	1200
varied_horn_behind	P11	4	200	100	1100
varied_horn_alongside	P12	5	150	100	1150

Total duration = (pulses × pulse duration) + ((pulses − 1) × pause). The two emergency-vehicle classes (ambulance siren, fire-truck siren) are shaded.

4.3 Frontend

Thefrontend isimplemented using Streamlit. It supports audio-file upload and laptop-microphone recording, and displays the predicted sound category and confidence. The pulse demonstration is carried out separately through the Wokwi simulation.

5. Model Evaluation

KARNA's machine-learningmodel was evaluated on a multi-class environmental sound classification task involving 12 sound categories, including ambulance sirens, fire-truck sirens, traffic noise, and nine different horn scenarios. The evaluation was performed using both mono-audio and stereo-audio model configurations. The figures below are baseline results from the current model version.

5.1 Dataset and training setup

The dataset contains1,132 original audio clips across 12 classes. Horn clips are split by horn scenario. A traffic-augmented version of the data adds 932 entries, and both versions are divided into training, validation and test sets as shown below.

Dataset version	Training	Validation	Test	Total
Clean (original) clips	737	197	198	1,132
Traffic-augmented entries	597	167	168	932

- **Optimizer.** AdamW, with a maximum of 35 epochs and early stopping.
- **Noise augmentation.** Traffic augmentation is explicitly incorporated into the emergency training scripts.
- **Evaluation reports.** Clean and traffic-noisy evaluation reports are produced, each with per-class F1 scores and confusion matrices.

The mono model achieved 75.13% validation accuracy with a macro F1-score of 67.20%. On the clean test dataset, it achieved 61.11% accuracy and a macro F1-score of 41.85%. Under noisy test conditions, its accuracy was 52.38%, with an emergency-class macro F1-score of 36.09%.

The stereo model achieved 74.62% validation accuracy with a macro F1-score of 66.49%. On the clean test dataset, it achieved 61.11% accuracy and a macro F1-score of 42.49%. Under noisy test conditions, it achieved 54.17% accuracy, with an emergency-class macro F1-score of 39.66%.

The comparison indicates that stereo input provided a modest improvement in performance under noisy conditions. However, distinguishing between similar sound categories, particularly in noisy environments, remains a challenge. Further improvements in dataset diversity, model training, and noise robustness are required.

Evaluation note: These are multi-class evaluation results, not car-horn-only accuracy figures. The noisy emergency-class macro F1 calculation excludes the traffic-noise class because it had zero samples in the noisy test set.

6. Functional Demonstration Flow

The current prototype can be demonstrated in the following sequence:

- **Start the Streamlit application.** Open the frontend used for the KARNA prototype.
- **Provide audio.** Upload an audio file or record a sound through the laptop microphone.
- **Run classification.** The trained model processes the audio and produces a predicted sound category.
- **View the result.** The frontend displays the predicted sound and confidence.
- **Generate the pulse pattern.** The prediction is passed to the pulse-mapping logic.
- **Observe the simulation.** In Wokwi, the LED blinks the same number of times as the mapped vibration pulses.

7. Current Scope and Limitations

Working in the current MVP	Not part of the current MVP
• Audio-file input • Laptop-microphone recording • ML sound prediction • Predicted category and confidence display • Sound-to-pulse mapping • Wokwi LED pulse demonstration	• A physically built wristband • A connected vibration motor • A claim of real-world haptic delivery to a wearer

The Wokwi LED is used only to make the pulse logic visible during demonstration. It is not the intended user-facing alert mechanism. The intended product is a wearable haptic wristband that would communicate the mapped patterns through vibration.

8. Next Development Stage

The next stage is to deploy the model on the ESP32 for on-device inference, integrate the validated sound-to-pulse workflow with the intended wristband hardware and vibration motor, and then test end-to-end performance with real users and varied acoustic conditions. Model accuracy and robustness, especially in noisy environments, should also be improved and re-evaluated before making reliability claims.

9. MVP Outcome

KARNA currently demonstrates a functional software prototype of its core concept: audio input → machine-learning sound classification → prediction display → haptic pulse mapping → simulated pulse output. The Wokwi LED provides a visible demonstration of the pulse sequence while the physical wristband remains a future hardware implementation.

Model evaluation figures are reproduced as recorded during development.

Demonstration Screenshots

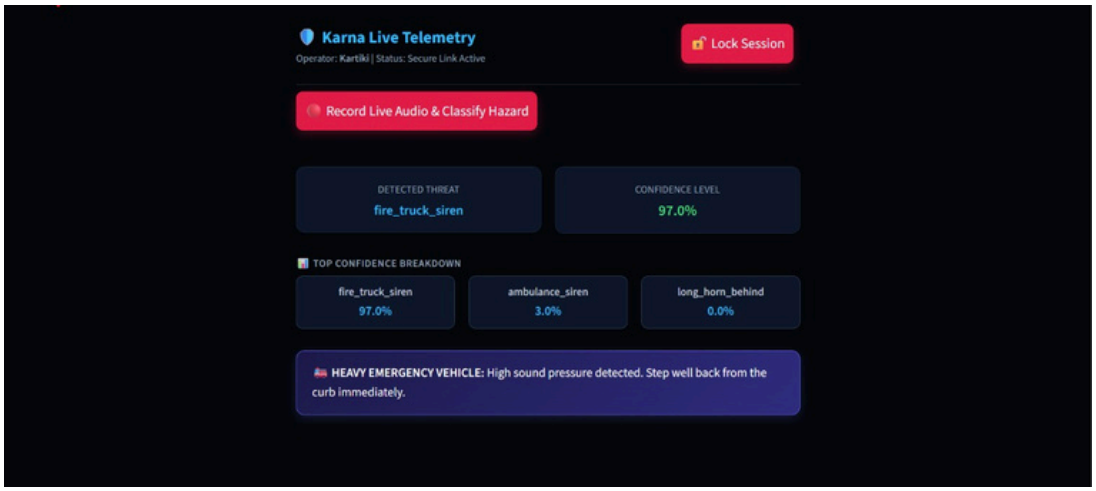


Figure 1. Streamlit frontend (Karna Live Telemetry): detected class ***fire_truck_siren*** at 97.0% confidence, with the top-confidence breakdown and on-screen alert.

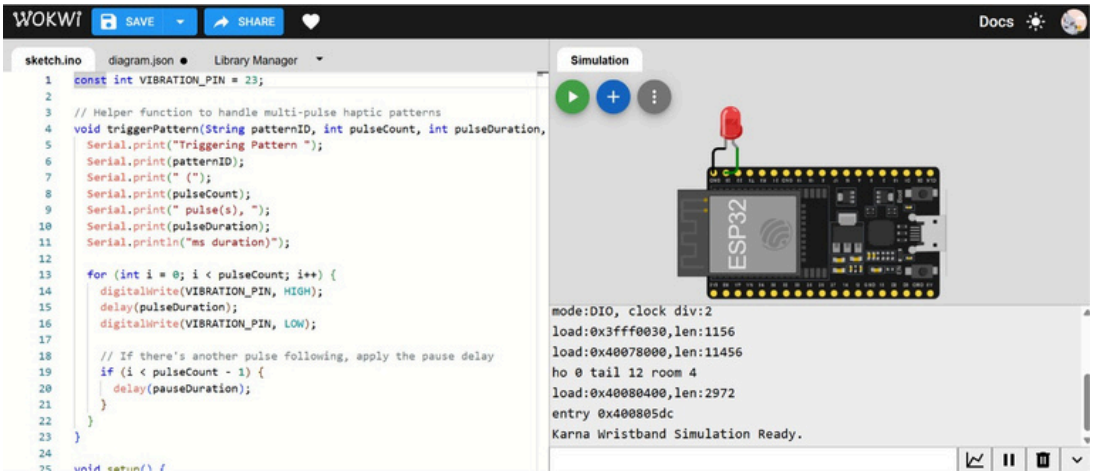


Figure 2. Wokwi ESP32 simulation: the triggerPattern function blinks the LED on pin 23 once per pulse and applies the pause delay between pulses; the serial monitor shows “Karna Wristband Simulation Ready.” The LED stands in for the intended wristband's vibration motor.